SBP Tugas 2

No.

Date

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R7B

No	Mhs	Minat	Asal Pendidikan	Gender	Hobi	Hasil
1	A	Hardware	SMK komputer	L	Non IT	TI
2	B	Hardware	SMK komputer	L	IT	TI
3	C	Umum	SMK Komputer	L	Non IT	SI
4	D	Software	SMA Umum	L	Non IT	SI
5	E	Software	SMK Teknik	P	Non IT	SI
6	F	Software	SMK Teknik	P	IT	SI
7	G	Umum	SMK Teknik	P	IT	SI
8	H	Hardware	SMA Umum	L	Non IT	TI
9	I	Hardware	SMK Teknik	P	Non IT	SI
10	J	Software	SMA Umum	P	Non IT	SI
11	K	Hardware	SMK Umum	P	IT	SI
12	L	Umum	SMA Umum	L	IT	SI
13	M	Umum	SMK Komputer	P	Non IT	SI
14	N	Software	SMA Umum	L	IT	TI

Data Awal

TI = 4 , SI = 10

* Entropy Awal

$$\text{Entropy}(S) = -P_1 \log_2(P_1) - P_2 \log_2(P_2)$$

$$P(TI) = \frac{4}{14} = 0,2857$$

$$P(SI) = \frac{10}{14} = 0,7143$$

$$\log_2(4/14) = \log_2(0,2857) = -1,8074$$

$$\log_2(10/14) = \log_2(0,7143) = -0,4854$$

$$\text{Entropy}(\text{Hasil}) = -(4/14 \log_2(4/14)) - (10/14 \log_2(10/14))$$

$$= -(0,2857 \cdot (-1,8074)) - (0,7143 \cdot (-0,4857))$$

$$= -(-0,5160) - (-0,3467)$$

$$= 0,5160 + 0,3467$$

$$= 0,8627 //$$

Data Minat

	Total	TI	SI
Hardwere	5	3	2
Software	5	1	4
Umum	4	0	4

$\Delta -\left(\frac{3}{5}\log_2\frac{3}{5}\right)-\left(\frac{2}{5}\log_2\frac{2}{5}\right) : 0,9710$

$\Delta -\left(\frac{1}{5}\log_2\frac{1}{5}\right)-\left(\frac{4}{5}\log_2\frac{4}{5}\right) : 0,7219$

$\Delta -\left(\frac{0}{4}\log_2\frac{0}{4}\right)-\left(\frac{4}{4}\log_2\frac{4}{4}\right) = 0,0000$

Entropy (Minat) $= (5/14 . 0,9710) + (5/14 . 0,7219) + (4/14 . 0)$
$= (0,3571 . 0,9710) + (0,3571 . 0,7219) + (0,2857 . 0)$
$= (0,3468 + 0,2578 + 0)$
$= 0,6046$//

Gain (Minat) : Entropy (Hasil) − Entropy (Minat)
$= 0,8622 - 0,6046$
$= 0,2581$//

Data Asal Pendidikan

	Total	TI	SI
SMK Komputer	4	2	2
SMA Umum	6	2	4
SMK Teknik	4	0	4

$\Delta -(2/4\log_2 2/4)-(2/4\log_2 2/4) = 1,0000$
$\Delta -(2/6\log_2 2/6)-(4/6\log_2 4/6) = 0,9183$
$\Delta -(0/4\log_2 0/4)-(4/4\log_2 4/4) = 0,0000$
Entrop (AP) $= (4/14 . 1) + (6/14 . 0,9183) + (4/14 . 0)$
$= (0,2857 . 1) + (0,4286 . 0,9183) + (0,2857 . 0)$
$= 0,2857 + 0,3936$
$= 0,6793$//
Gain (AP) $= 0,8627 - 0,6793 = 0,1834$//

Data Gender

	Total	TI	SI
L	7	4	3
P	7	0	7

$\Delta - (4/7 \log_2 4/7) - (3/7 \log_2 3/7) = 0,9852$

$\Delta - (0/7 \log_2 0/7) - (7/7 \log_2 7/7) = 0,0000$

Entropy (Gender) $= (7/14 . 0,9852) + (7/14 . 0)$

$= (0,5 . 0,9852) + (0,5 . 0)$

$= 0,4926$ //

Gain (Gender) $= 0,8627 - 0.4926 = 0,3701$ //

Data Hobi:

	Total	TI	SI
Non IT	8	2	6
IT	6	2	4

$\Delta - (2/8 \log_2 2/8) - (6/8 \log_2 6/8) = 0,8113$

$\Delta - (2/6 \log_2 2/6) - (4/6 \log_2 4/6) = 0,9183$

Entropy (Hobi) $= (8/14 . 0,8113) + (6/14 . 0,9183)$

$= (0,5714 . 0,8113) + (0,4286 . 0,9183)$

$= 0,4638 + 0,3936$

$= 0,8574$ //

Gain (Hobi) $= 0.8627 - 0,8574 = 0,0053$ //

∴ Gain Gender tertinggi. (L)

Entropy (SL) $= 0,9852$ //

Data Minat pada SL

	Total	TI	SI	Entropy (SV)
Hardware	3	3	0	0,0000
Software	2	1	1	1,0000
Umum	2	0	2	0,0000

Entropy (Minat) $= (3/7 . 0) + (2/7 . 1) + (2/7 . 0)$

$= (0,4285 . 0) + (0,2857 . 1) + (0,2857 . 0)$

$= ,2857$ //

Gain (Minat) $= 0,9852 - 0,2857 = 0,6995$ //

Data Asal Pendidikan pada SL

	Total	TI	SI	Entropy (SV)
SMK komputer	3	2	1	0,9183
SMA Umum	4	2	2	1,0000

Entropy (AP) = $(3/7 \cdot 0,9183) + (4/7 \cdot 1)$
= $(0,4286 \cdot 0,9183) + (0,5714 \cdot 1)$
= $0,3936 + 0,5714$
= $0,9650$ //

Gain (AP) : $0,9852 - 0,9650 = 0,0202$ //

Data Hobi pada SL

	Total	TI	SI	Entropy (SV)
Non IT	4	2	2	1,0000
IT	3	2	1	0,9183

Entropy (Hobi) = $(4/7 \cdot 1) + (3/7 \cdot 0,9183)$
= $(0,5714 \cdot 1) + (0,4286 \cdot 0,9183)$
= $0,5714 + 0,3936$
= $0,9650$ //

Gain (Hobi) = $0,9852 - 0,9650 = 0,0202$ //

∴ Gain Minat tertinggi. (software)

Entropy (SL, software) = 1,0000 //

Data Hobi pada SL, software

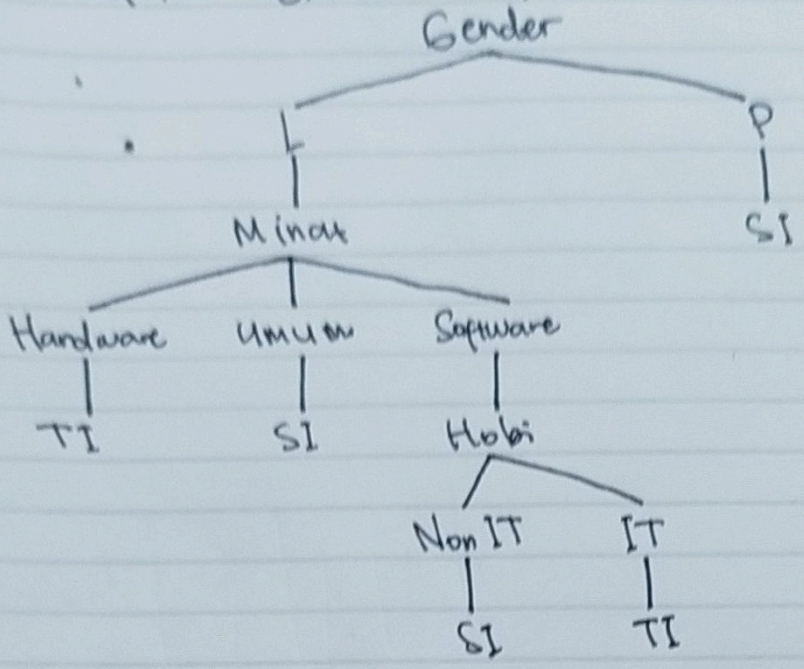
	Total	TI	SI	Entropy (SV)
Non IT	1	0	1	0,0000
IT	1	1	0	0,0000

Entropy (Hobi) = $(1/2 \cdot 0) + (1/2 \cdot 0)$
= 0 //

Gain (Hobi) = $1 - 0 = 1$ //

∴ Gain Hobi tertinggi.

Pohon Keputusan (Decision Tree)



Aturan Keputusan (Rule Base)

1. If Gender = P → THEN Hasil = SI.
2. If Gender = L && Minat = Hardware → THEN Hasil = TI.
3. If Gender = L && Minat = Umum → THEN Hasil = SI.
4. If Gender = L && Minat = Software && Hobi = Non IT → THEN Hasil = SI.
5. If Gender = L && Minat = Software && Hobi = IT → THEN Hasil = TI.